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CS499

Artifact 1: Travlr Getaways

Software Design and Engineering

Description of the Artifact

The first artifact I selected for my ePortfolio is my Travlr Getaways project from CS465. This was a full-stack web application that I worked on using technologies including Angular, Express, Node.js, and MongoDB. The project gave me experience working with different parts of a web application and understanding how they work together.

I selected this project because it represents one of the areas of computer science that I am most interested in. Web development was one of the areas that stood out to me during the program, and the Travlr project gave me experience working with both the front end and back end of an application.

Why I Selected This Artifact

I chose Travlr Getaways because it shows my experience with software development and designing an application that has multiple parts. I had to understand how the different components worked together instead of only writing one small program.

The project also gave me experience with Angular, Express, Node.js, MongoDB, routing, and the overall structure of a full-stack application. These are skills that I can use as a foundation for future web development work.

For the enhancement, I focused on improving the administrative side of the application. One of the areas I wanted to improve was making it easier to work with and search for trips. I also looked at ways to make the administrative functions more organized and protect editing and saving functionality.

What I Learned From the Enhancement

Going back to an older project and trying to improve it was different from creating something from scratch. I had to first understand how the existing application worked before deciding what needed to be changed.

This helped me understand why software design is important. If the different parts of an application are not organized well, adding new features can become harder. I also learned that changes made to one part of a full-stack application can affect other parts of the application.

The enhancement helped me practice reading existing code, understanding how components connect, and thinking about how a user would actually use the application. It also gave me more experience thinking about security and protecting administrative functionality.

Challenges and Feedback

One challenge with this project was understanding the different technologies used together. Angular, Node.js, Express, and MongoDB each have their own purpose, and it took time to understand how they connected.

I also had to troubleshoot problems while working on the project. Working through those problems helped me become more comfortable with finding errors and figuring out where something was going wrong.

Feedback throughout the CS499 process also helped me look at the project differently. Instead of only asking whether the program worked, I started thinking about whether the project was organized, understandable, secure, and useful.

Course Outcomes

This artifact supports several of the CS499 course outcomes. It demonstrates my ability to design and evaluate a computing solution, use different tools to implement a solution, and think about security when making changes to software.

It also helped me practice communicating technical information in a way that another person could understand. The project is especially connected to software engineering and development because I had to work with an existing application and consider how to expand its functionality without breaking the rest of the project.